G V S S Krishna Mowli

M.Tech Systems and Control - TA

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Degree	University/Institute	Year	CGPA/Marks(%)
M.Tech Systems and Control - TA	IIT Hyderabad	2027	9
B.Tech	National Institute of Technology Tirucchirapalli	2025	8.32
XII (CBSE)	Krishnaswamy Vidyaniketan	2021	92.60%
X (CBSE)	Krishnaswamy Vidyaniketan	2019	89.20%

SCHOLASTIC ACHIEVEMENTS

Secured National Ranks 54 and 53 in **NISO** conducted by **EduHeal Foundation** in 2017 and 2018, respectively .
Secured National Ranks 56 and 57 in **NIMO** conducted by **EduHeal Foundation** in 2017 and 2018; **International Rank 151** after qualifying for Level 2 in 2017.
Earned Gold Medals (Medal of Honor) in **INEO** and **SIMO** in 2014.
Ranked within the top 2,500 among 164,962 applicants in **TS EAMCET Engineering 2021** (~top 1.5%).
Selected as the sole student from Alpha Wisdom Vidyashram, Tiruchirappalli, for the **state-level Spelling Bee finals** in 2017.

SKILLS

Programming & Software: Python, MATLAB, ROS, Gazebo, C, Verilog, SQL (Basic), Git, GitHub, Linux, LaTeX
Robotics, AI & Machine Learning: Multi-Agent Systems, Swarm Robotics, Reinforcement Learning, Federated Learning, PPO.
Control, Optimization & Systems: Control Theory, State- Space Modeling, PID Control, LQR, MPC, Optimal Control, Robust Control, Adaptive Control, Nonlinear Control, Distributed Control, State Estimation, Kalman Filter, Lyapunov Stability, System Identification, Convex Optimization, QP, SOCP, LMI, Dynamic Programming.
Embedded Systems & IoT: Embedded Systems, Arduino, Raspberry Pi, IoT, Sensors, Actuators, ADC/DAC, PWM, UART, I^2C , CAN.
VLSI, FPGA & Semiconductor: Digital VLSI, Analog IC Design, CMOS, PDSOI, FPGA, RTL Design, Verilog, Cadence Virtuoso, Vivado, Sentaurus TCAD, Logic Synthesis (Basic).
Quantitative Methods: Probability, Stochastic Processes, Convex Optimization (QP/SOCP), Dynamic Programming, Monte Carlo Simulation
Machine Learning & Data Analytics: Data Preprocessing, Feature Engineering, Supervised Learning, Classification, Model Training, Validation & Evaluation, PyTorch, Scikit-learn

RELEVANT COURSES

Multi-Agent Systems | Reinforcement Learning | Machine Learning & Applications | Fundamentals of Robotics
Digital Signal Processing | Signals & Systems | Digital Image Processing | Real Analysis & Probability Theory
Optimal Control | State Estimation | Introduction to State-Space | Convex Optimization | Advanced Control Lab |
Matrix Theory | Random Variables & Stochastic Processes | Advanced Stochastic Processes
Network Analysis & Synthesis | Digital Circuits & Systems | Electronic Circuits | Analog Integrated Circuits | VLSI Systems
VLSI Design: RTL to GDS | Embedded Systems | Microprocessors & Microcontrollers | Semiconductor Physics & Devices
Wireless Communication | Digital Signal Processors & Applications | Automotive Communication & Sensing

PROJECTS

Ultra-Low-Power MPPT Architecture for Battery-less IoT Energy Harvesting Jan– May 2025
Cadence Virtuoso — Analog IC Design — Energy Harvesting

- Architected an **ultra- low- power Time- to- Voltage Boost Converter** in **Cadence Virtuoso** using **UMC 45nm CMOS** to maximize photovoltaic energy harvesting across battery-less wireless IoT sensor nodes.
- Designed an **adaptive step- size Hill- Climbing controller** with **VCO- based perturbation logic**, substantially accelerating dynamic tracking speed while suppressing parasitic steady-state power oscillations.
- Optimized front- end **quiescent current dissipation** and verified layout parasitics, achieving a verified **99.9985% MPPT tracking efficiency** across broad dynamic irradiance operating ranges.

Ultra-Low-Power MPPT Architecture for Battery-less IoT Energy Harvesting (UMC 45nm CMOS) Jan– May 2025
Cadence Virtuoso — Analog IC Design — Energy Harvesting

- Architected a Time-to-Voltage Controlled Boost Converter enabling ultra-low-power maximum power point tracking (MPPT) for photovoltaic energy harvesting in battery-less IoT sensor nodes.
- Designed an adaptive step-size Hill-Climbing controller employing VCO-based perturbation control, significantly improving convergence speed while minimizing steady-state oscillations.
- Optimized the analog front-end for minimal quiescent current, making the architecture suitable for self-powered edge devices.
- Verified the complete design through post-layout Cadence simulations, achieving 99.9985% MPPT tracking efficiency under varying irradiance conditions.

Fault-Tolerant Turbo Decoder for SRAM-Based FPGAs Apr–May 2024
Verilog — FPGA — Digital VLSI — Fault Tolerance — Vivado

- Architected a radiation-tolerant **Turbo Decoder architecture** on **SRAM- based FPGAs** using **Xilinx Vivado**, mitigating critical **Single Event Upsets (SEUs)** across configuration memory and user logic.
- Implemented a **Duplication- With- Comparison (DWC)** framework tracking **LLR confidence metrics** and sign changes, detecting internal data faults with significantly lower area overhead than **TMR**.
- Executed rigorous **FPGA fault- injection simulations** and reliability benchmarks, demonstrating robust hardware fault tolerance targeted for mission-critical aerospace communication links

Fault-Tolerant Turbo Decoder for SRAM-Based FPGAs Apr–May 2024
Verilog — FPGA — Digital VLSI — Fault Tolerance — Vivado

- Designed and analyzed a fault-tolerant Turbo Decoder architecture for SRAM-based FPGAs capable of mitigating Single Event Upsets (SEUs) in both configuration and user memories.
- Implemented a Duplication-With-Comparison (DWC) architecture that detects decoder inconsistencies using decoded outputs, Log-Likelihood Ratio (LLR) confidence metrics, and sign-change monitoring.
- Evaluated architectural trade-offs between DWC and Triple Modular Redundancy (TMR), demon strating significantly lower hardware overhead while maintaining high fault tolerance.
- Investigated FPGA fault-injection methodologies and reliability-aware digital design techniques for radiation-prone aerospace and satellite communication systems.

Resilient Multi-Agent Coordination using Federated Reinforcement Learning Jan– May 2026
Reinforcement Learning — Robotics — Distributed Control

- Developed a decentralized **multi- agent coordination framework** coupling **Federated PPO** with **Age- of- Information (AoI)** state prediction to ensure resilient swarm consensus over lossy channels.
- Integrated **Distributed Control Barrier Functions (DCBFs)** with differential games, guaranteeing provable **collision avoidance** and robust formation control for heterogeneous robotic swarms.

Resilient Multi-Agent Coordination using Federated Reinforcement Learning Jan– May 2026

Reinforcement Learning — Robotics — Distributed Control

- Developed a decentralized multi-agent coordination framework integrating Federated PPO with Age of-Information-aware state prediction.
- Combined Distributed Control Barrier Functions (DCBFs) with differential-game-based safety control for provable collision avoidance.
- Designed communication-efficient learning strategies for heterogeneous robotic swarms operating un der delayed and unreliable networks.
- Demonstrated robust formation control and scalable decentralized decision making.

Tax-Aware Portfolio Optimization under Tax and Transaction-Cost & E.S.G. Constraints Portfolio Optimization Jan– Apr 2026

SOCP/QP — Python — Financial Engineering

- Formulated a convex portfolio optimization pipeline in **Python** integrating capital gains tax schedules, nonlinear transaction fees, **wash-sale constraints**, and explicit **ESG sustainability** metrics.
- Engineered a sequential **SOCP and Quadratic Programming (QP)** solver for tax-efficient rebalancing and sign-constrained allocation, consistently converging to globally optimal risk-diversified assets.

Tax-Aware Portfolio Optimization under Tax and Transaction-Cost & E.S.G. Constraints Portfolio Optimization Jan– Apr 2026

SOCP/QP — Python — Financial Engineering

- Designed a comprehensive portfolio optimization framework integrating capital gains taxation, trans action costs, wash-sale constraints, and ESG objectives into a unified convex optimization model.
- Formulated the investment problem as a sequence of Second-Order Cone Programming (SOCP) and Quadratic Programming (QP) problems to efficiently compute globally optimal portfolios.
- Implemented a two-stage optimization strategy for tax-efficient rebalancing, multi-account asset lo cation, and sign-constrained refinement while maintaining portfolio diversification.
- Evaluated the framework across multiple investment scenarios, demonstrating effective balancing of risk, return, taxation, trading costs, and sustainability objectives.

Direct Data-Driven Optimal Control of Dynamic Systems LMI Optimization Mar– Apr 2026

Data-Driven Control — MATLAB

- Synthesized a direct **data-driven LQR controller** eliminating parametric system identification, using **Lyapunov-based LMIs** and **Schur complements** to guarantee mean-square closed-loop stability.
- Evaluated controller robustness across **500 Monte-Carlo simulation trials** in **MATLAB**, matching model-based **NSMLQR benchmarks** within an ultra-tight **2% relative error bound**.

Direct Data-Driven Optimal Control of Dynamic Systems LMI Optimization Mar– Apr 2026

Data-Driven Control — MATLAB

- Designed a direct data-driven LQR controller eliminating the requirement for explicit plant identification.
- Formulated Lyapunov-based Linear Matrix Inequalities (LMIs) with Schur-complement transformations to guarantee mean-square closed-loop stability under stochastic disturbances.
- Evaluated robustness using 500 Monte-Carlo simulations, achieving control performance within 2% of the model-based NSMLQR benchmark.
- Demonstrated the practical applicability of optimization-based control for uncertain dynamical systems.

Modeling & Control of Electromechanical Systems System Modeling Jan– Apr 2026

State Space — MATLAB/Simulink

- Derived first-principles equations and nonlinear **state-space representations** in **MATLAB/Simulink** for Linear Motion Carts, Rotary Inverted Pendulums, and complex Gantry Cart systems.
- Synthesized classical and modern **trajectory-tracking state-feedback controllers**, validating closed-loop disturbance rejection and transient stability across physical laboratory setups.

Modeling & Control of Electromechanical Systems System Modeling Jan– Apr 2026

State Space — MATLAB/Simulink

- Developed first-principles mathematical models for Linear Motion Cart, Rotary Inverted Pendulum, and Gantry Rotary Pendulum systems.
- Derived nonlinear and linearized state-space representations for controller synthesis and stability analysis.
- Designed trajectory-tracking and stabilization controllers using classical and modern control techniques.
- Validated closed-loop performance through simulation and experimental laboratory investigations.

Optimal Control Analysis using Finite & Infinite Horizon LQR MATLAB Nov 2025

Control Theory — Riccati Equations

- Formulated finite and infinite-horizon **Linear Quadratic Regulators (LQR)** by evaluating continuous **Differential and Algebraic Riccati Equations**, analytically deriving optimal state and co-state paths.
- Built a modular **MATLAB simulation framework** to analyze dynamic performance trade- offs between tracking speed, state convergence rates, and total closed-loop actuator control effort.

Optimal Control Analysis using Finite & Infinite Horizon LQR MATLAB Nov 2025

Control Theory — Riccati Equations

- Designed finite- and infinite-horizon Linear Quadratic Regulators (LQR) by solving Differential and Algebraic Riccati Equations.
- Analytically derived optimal state, input, and costate trajectories while validating theoretical optimality conditions.
- Compared finite- and infinite-horizon formulations to study stability, convergence, and control-energy trade-offs.
- Implemented a modular MATLAB simulation framework for optimal controller synthesis and performance evaluation.

Constrained Optimal Control using Dynamic Programming Oct 2025

MATLAB — Dynamic Programming — Bellman Optimization

- Implemented a Bellman-based **Dynamic Programming solver** in **MATLAB** for finite-state systems, computing globally optimal control sequences under hard state and actuator input constraints.
- Formulated recursive **cost- to- go equations** via backward induction, verifying policy convergence against analytical optimality benchmarks to establish an extensible numerical control engine.

Constrained Optimal Control using Dynamic Programming Oct 2025

MATLAB — Dynamic Programming — Bellman Optimization

- Implemented a Bellman-based Dynamic Programming framework for finite-state constrained optimal control problems.
- Formulated recursive cost-to-go equations enabling globally optimal policy generation under state and input constraints.
- Generated optimal control policies and state trajectories through backward recursion while validating convergence against analytical benchmarks.
- Built a reusable MATLAB framework capable of solving generalized constrained optimal control problems.

Distributed Consensus in Multi-Robot Systems with Collision & Obstacle Avoidance - Ongoing Thesis Jun 26 - Present

- **Developing a decentralized multi-robot consensus framework** using **weighted graph interactions and Laplacian-based coordination** for distributed position agreement and collective field estimation.
- **Formulating a probabilistic collision-avoidance guarantee** using **Control Barrier Functions,** process-noise modeling, and Cantelli's inequality to bound pairwise safety violations.
- **Deriving pointwise and mission-horizon safety bounds**, introducing a **fixed minimum clearance** and extending the guarantee from individual robot pairs to the complete multi-robot team.
- **Implementing gradient-based field estimation** with **basis functions** and integrating the estimated field gradient with distributed coordination for decentralized multi-robot motion.
- **Validating** the proposed **probabilistic guarantees through Monte Carlo simulation**, comparing theoretical bounds with empirical collision probabilities across safety margins and mission horizons.

Distributed Consensus in Multi-Robot Systems with Collision & Obstacle Avoidance - Ongoing Thesis Jun 26-Present

- **Developing a decentralized multi-robot consensus framework** using **weighted graph theory and Laplacian-based coordination** for position agreement and distributed field-parameter estimation.
- **Formulating probabilistic collision-avoidance guarantees** using **Control Barrier Functions and Cantelli's inequality**, deriving pointwise and mission-horizon bounds with a **fixed safety margin**.
- **Implementing gradient-descent field estimation with basis functions** and validating the proposed safety guarantees through **Monte Carlo simulations** against empirical collision probabilities.

Distributed Consensus in Multi-Robot Systems with Collision & Obstacle Avoidance - Ongoing Thesis Jun 26-Present

- Developing a **decentralized multi-robot consensus framework** using **weighted graphs & Laplacian coordination**.
- **Implementing gradient-descent field estimation** using basis functions for distributed parameter estimation.
- Formulating **probabilistic collision-avoidance guarantees** using **Control Barrier Functions** and Cantelli's inequality.
- **Validating pointwise and mission-horizon safety bounds** through **Monte Carlo simulations** with fixed safety margins.

Customer Churn Prediction & Segmentation | GitHub · Live Dashboard Sep-Oct 2026
| Python, SQL, scikit-learn, Streamlit |

- Built an end-to-end churn prediction and customer segmentation pipeline on 7,000+ telco customers, identifying a high-risk segment responsible for **68% of lost monthly revenue** ($1.7M annualized) from just 28% of customers.
- Engineered a leakage-safe ML pipeline (logistic regression, random forest, gradient boosting) achieving **0.839 ROC-AUC** (95% CI 0.815–0.860); targeting the top 20% by risk score captures **50% of churners at 2.5x lift** over random outreach.
- Applied K-Means clustering to segment customers into four actionable groups without using the churn label, then validated separation using bootstrap confidence intervals and two-proportion hypothesis tests.
- Quantified churn risk drivers using Kaplan-Meier survival curves and a stratified Cox proportional hazards model, finding protective service add-ons cut churn hazard by 34–47%.
- Wrote 8 SQL queries (joins, window functions, CTEs) against a normalized SQLite schema to surface revenue-at-risk and retention patterns for stakeholder reporting.
- Delivered an interactive Streamlit dashboard and executive summary translating model output into three segment-specific, ROI-justified retention recommendations.

Customer Churn Prediction & Segmentation | GitHub · Live Dashboard Sep-Oct 2026
| Python, SQL, scikit-learn, Streamlit |

- Built an end-to-end churn prediction pipeline (SQL, leakage-safe ML) on 7K+ customers, achieving **0.839 ROC-AUC** with the top 20% risk-ranked customers capturing **50% of churners at 2.5x lift**.
- Segmented customers via K-Means into 4 groups, identifying one segment responsible for **68% of lost revenue** from just 28% of customers.
- Validated findings with bootstrap confidence intervals and Cox survival analysis.

Customer Churn Prediction & Segmentation | GitHub · Live Dashboard Sep-Oct 2026
| Python, SQL, scikit-learn, Streamlit |

- Built an end-to-end churn prediction pipeline (SQL, leakage-safe ML, K-Means segmentation) on 7K+ customers, achieving **0.839 ROC-AUC** and identifying a segment responsible for **68% of lost revenue** from just 28% of customers.
- Validated results with bootstrap confidence intervals and Cox survival analysis, then delivered them via an interactive Streamlit dashboard — the top 20% risk-ranked customers capture **50% of churners at 2.5x lift** over random targeting.

GenAI-Enabled Task Planning and Dynamic Allocation for Heterogeneous Autonomous Agents Aug-Sep 2026
| Python, LLMs, Optimization, Multi-Agent Simulation|

- Designed a Generative AI-based planning framework to translate natural-language missions into structured subtasks with dependencies, resource requirements, and execution constraints.
- Implemented capability-aware multi-agent task allocation using optimization-based assignment, integrating deterministic plan validation, resource constraints, and collision-aware path planning.
- Developed an event-driven replanning mechanism to recover from agent failures and changing environmental conditions, with quantitative evaluation against rule-based baselines.
- Built a reproducible simulation and testing framework to measure mission completion rate, execution time, allocation cost, planning latency, and safety violations across controlled scenarios.

GenAI-Enabled Task Planning and Dynamic Allocation for Heterogeneous Autonomous Agents Aug-Sep 2026
| Python, LLMs, Optimization, Multi-Agent Simulation |

- Developed a Generative AI-based planning framework to convert **natural-language missions** into structured subtasks with dependencies, resource requirements, and execution constraints.
- Implemented **optimization-based task allocation** for heterogeneous agents, integrating deterministic plan validation, resource constraints, and **collision-aware path planning**.
- Built an **event-driven dynamic replanning** and simulation framework to **handle agent failures** and environmental changes, evaluating mission completion, allocation cost, execution time, and safety violations against baseline methods.

EXTRACURRICULAR

- Received professional training from TENVIC's elite coaches in Cricket, Table Tennis, and Basketball, strengthening teamwork, discipline, and competitive performance.
- Participated in State-Level Hockey Championships, demonstrating teamwork, discipline, and competitive spirit.
- Trained under **TENVIC coaches** across **Cricket, Table Tennis, and Basketball** to build discipline and competitive drive.

INTERNSHIPS

FPGA & DSP Design Research Intern | NIT Tiruchirappalli Guide: Dr. G. Lakshminarayanan | Dec 2023– Jan 2024

- Designed and verified a **4-level pipelined architecture and Braun multiplier** for implementation on an **Artix-7 FPGA using Xilinx Vivado**.
- Developed **Verilog-based DSP processor modules**, including program memory, instruction-fetch logic and stack operations, with read/write control for memory components.

- Built and validated RTL designs through **simulation and functional testbenches**, gaining practical experience in FPGA-based digital design and DSP hardware implementation.

FPGA & DSP Design Research Intern | NIT Tiruchirappalli Guide: Dr. G. Lakshminarayanan | Dec 2023– Jan 2024

- Designed and verified a **4-level pipelined architecture** and **Braun multiplier** for hardware implementation on a **Xilinx Artix-7 FPGA**, executing comprehensive synthesis, placement, and timing closure in **Vivado**.
- Developed modular **Verilog DSP processor cores**, integrating instruction-fetch logic, **program memory interfaces**, stack execution pipelines, and synchronous read/write control registers for memory subsystems.
- Constructed self-checking **RTL behavioral testbenches** and functional simulation suites in **Vivado**, validating cycle-accurate processor execution and eliminating internal race conditions before hardware sign-off.

PDSOI–MOS Design Research Intern | IIT Roorkee Guide: Dr. B. Anand Bulusu | Jun–Jul 2024.

- Studied **Partially Depleted SOI (PDSOI) MOSFETs**, focusing on floating-body and self-heating effects and their impact on device characteristics.
- Performed **I–V and C–V device characterization using Sentaurus TCAD**, including modelling of floating-body potential and extraction of simulation data.
- Designed and simulated **CMOS inverter, ring oscillator, operational amplifier, comparator and common-source amplifier** circuits, comparing PDSOI behaviour with bulk MOSFET implementations.

PDSOI–MOS Design Research Intern | IIT Roorkee Guide: Dr. B. Anand Bulusu | Jun–Jul 2024.

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- Performed **I–V & C–V device characterization in Sentaurus TCAD** & modelled floating-body potential & extraction of simulation data.
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IoT Systems Intern — RINEX | IITR & IITBbh Mar– May 2022

- **Developed IoT prototypes** for environmental sensing & device control using **Arduino, Adafruit, and MIT App Inventor**.
- **Built an Arduino-based RC car** implementing **forward, reverse, left, right, and stop** controls.
- **Developed a Telegram-integrated IoT chatbot** using **Python and Adafruit IO** for remote light control; explored **Speech-to-Text and ThingSpeak** for voice control and data visualization.

IoT Systems Intern — RINEX | IITR & IITbbh Mar– May 2022

- Developed foundational **IoT prototypes** involving environmental sensing, mobile control and cloud-connected device interaction using **Arduino, Adafruit IO and MIT App Inventor**.
- Built an **Arduino-based command-operated RC car**, implementing directional control including forward, reverse, left, right and stop operations.
- Developed a **Telegram-integrated IoT chatbot** for remotely controlling a connected light through **Python and Adafruit IO**, gaining hands-on experience with IoT communication and automation.
- Explored **voice-based device control and IoT data visualization** using MIT App Inventor, Speech-to-Text and ThingSpeak.

Image Processing & CNN Internship | NIT Tiruchirapalli Guide: Dr. P. Palanisamy | Dec 2022– Jan 2023

- Explored image classification using **Python, PyTorch, FER2013, CK+, VGG19, and ResNet18**.
- Implemented computer vision workflows for preprocessing, augmentation, training, validation, and evaluation.
- Engineered HOG, LBP, and colour-histogram features and evaluated **SVM and Random Forest** classifiers.
- Analyzed image classification through **feature engineering, CNN architecture, and model evaluation**.

Image Processing & CNN Internship | *NIT Tiruchirapalli* NITT (Guide: Dr. P. Palanisamy) | Dec 2022– Jan 2023

- Explored **image classification using Python and PyTorch**, working with facial-expression datasets such as **FER2013 and CK+** and studying CNN architectures including **VGG19 and ResNet18**.
- Developed familiarity with the end-to-end **computer vision pipeline**, including image preprocessing, dataset preparation, augmentation, model training, validation and performance evaluation.
- Implemented and experimented with **feature-based image classification** for an Autism Spectrum Disorder (ASD) dataset, extracting **HOG, LBP and colour-histogram features** and evaluating **SVM and Random Forest classifiers**.
- Gained practical exposure to **classification workflows, feature engineering, CNN architectures and model evaluation**, developing foundational understanding of how different image-classification approaches perform across datasets.

HOBBIES AND INTERESTS

Sports : Table Tennis, Cricket, Chess, Badminton, Basketball, Volleyball, Carroms
Reading: Diverse genres spanning fiction, philosophy, self-development, history, and Indian literature
Astrology: Independent study and research of astrology and horoscopes of several individuals
Indian Knowledge Systems: Exploring the **Vedas, Hindu philosophy, mythology, and spiritual literature**; regularly listening to talks and discourses by scholars
Financial Markets: Self-directed study and research on stock markets and investing
Cooking: Passion to learn different dishes and Occasional home cooking for family
Gaming: PC and mobile gaming, interest in multiplayer, competitive, and strategy-based games (big fan of Supercell)
Interests: Sports, chess, reading, Indian philosophy and literature, financial markets, astrology, cooking, and strategy gaming.
Table tennis, cricket, chess, badminton, reading across fiction, philosophy & history, and Indian Culture.
Financial markets and investing, Astrology, home cooking, and competitive PC/mobile strategy games.

CERTIFICATIONS

Semiconductor Specialization KAIST Coursera

1. Semiconductor Physics
2. Diode– PN Junction and Metal Semiconductor Contact
3. Transistor– Field Effect Transistor and Bipolar Junction Transistor

Matrix Algebra HKUST Coursera
Introduction to Basics Vibrations KAIST Coursera

POSITIONS OF RESPONSIBILITY

Core Member – Akshara Sanjeevini (Telugu Cultural Club) Jul 2026-present

- **Coordinated core-team planning** for the club's official inauguration; **organized event schedules**, handled guest hospitality, and facilitated the official name and logo launch during an invited Ashtavadhani scholar's campus visit.

Teaching Assistant – Control Systems | IIT Hyderabad Aug 2025–present

- **Guided students** in Control Systems through course assignments and academic activities, while supporting **examinations, invigilation, and evaluation** under the faculty instructor.

Placement Coordinator – Systems & Control | IIT Hyderabad Jun 2026–present

- **Coordinated placement activities** for the Systems & Control specialization, serving as the primary liaison among **recruiters, students, faculty, and the Institute Placement Office**.

Publicity Coordinator – Probe'23 | NIT Tiruchirappalli Jan-Mar 2023

- **Led outreach to 50+ colleges** through virtual and on-site campaigns for the ECE department's national-level symposium, contributing

to **1,500+ student footfall.**

Volunteer – Prayas, Rural Development Center (RDC) | IIT Hyderabad July 2026–present

- Volunteering with the Rural Development Center (RDC) under Prayas, with upcoming involvement in **planning and organizing community-development initiatives** for rural and nearby communities.